

Email Risk Analyzer

Phishing Analysis & Security Audit Tool

A Python-based Email Risk Analyzer that identifies common phishing indicators in email headers and body content using a rule-based scoring system.

Features

- Parses email headers and body content.
- Extracts From, Subject, Return-Path, and Authentication-Results.
- Checks SPF and DKIM authentication results.
- Detects Return-Path and sender-domain mismatches.
- Detects urgent and pressure-based language.
- Extracts HTTP/HTTPS links and compares their domains with the sender domain.
- Calculates a rule-based risk score from 0 to 100.
- Classifies emails as LOW, MEDIUM, or HIGH.
- Displays detected security indicators and their points.
- Includes safe and phishing-like synthetic test emails.

Project Structure

```
Email-Risk-Analyzer/  
■■■ email_risk_analyzer.py  
■■■ Phishing_Analysis_Report.pdf  
■■■ README.md
```

Technologies Used

- Python 3
- Python email module
- Python re module
- Python urllib.parse
- Regular expressions
- Rule-based security analysis

How the Analyzer Works

```
Email Input  
↓  
Parse Email Headers & Body  
↓  
Extract From / Subject / Return-Path  
↓  
Check SPF & DKIM  
↓  
Detect Urgent Language  
↓  
Analyze Links  
↓  
Check Domain Mismatches  
↓  
Calculate Risk Score  
↓  
LOW / MEDIUM / HIGH
```

Risk Scoring

Security Indicator	Points
Urgent or pressure-based language	+20
Suspicious/link domain mismatch	+20 to +25
SPF failure	+15
DKIM failure	+15
Return-Path mismatch	+10

Risk Levels: 0–39 = LOW 40–69 = MEDIUM 70–100 = HIGH

Requirements & Installation

Requirement: Python 3.x. No external Python packages are required for the current version.

Clone:

```
git clone https://github.com/YOUR-USERNAME/email-risk-analyzer.git
cd email-risk-analyzer
```

Run:

```
python email_risk_analyzer.py

# or
python3 email_risk_analyzer.py
```

Example Test Cases

Test 1 — Safe Synthetic Email

```
From: security@example.com
Subject: Routine security notice
SPF: pass
DKIM: pass
```

```
Expected:
Risk  : LOW
Score : 0/100
```

Test 2 — Synthetic Phishing-Like Email

```
From: security@example.com
Return-Path: alerts@other-example.com
SPF: fail
DKIM: fail
Subject: URGENT - Verify your account immediately
```

```
Expected:
Risk  : HIGH
Score : 80/100
```

Detected indicators: SPF failure (+15), DKIM failure (+15), Return-Path mismatch (+10), urgent language (+20), and link/domain mismatch (+20).

Example Output

```
=====
EMAIL SECURITY RISK ANALYZER
=====
From      : security@example.com
Subject   : URGENT - Verify your account immediately
Risk      : HIGH
Score     : 80/100

Findings:
- SPF failure (+15)
- DKIM failure (+15)
- Return-Path domain differs from From domain (+10)
- Urgent or pressure-based language detected (+20)
- Link domain differs from sender domain (+20)
```

Limitations

- Rule-based educational analyzer; it does not use machine learning or external threat intelligence.
- False positives and false negatives may occur.
- SPF/DKIM failures and domain mismatches are indicators, not proof of phishing.
- The score is a heuristic and should not be treated as a definitive security verdict.

- The supplied examples use synthetic/example domains and do not require real accounts or live links.

Ethical Use

Intended for cybersecurity education, phishing-awareness training, email-security demonstrations, academic projects, and defensive security analysis. Use only with data you are authorized to analyze.

Project Deliverables

```
email_risk_analyzer.py
Phishing_Analysis_Report.pdf
README.md
```

Future Improvements

Machine-learning classification; threat-intelligence integration; advanced URL analysis; domain reputation; redirect detection; attachment analysis; HTML analysis; DNS/domain-age checks; improved authentication parsing; GUI; web dashboard; JSON/CSV reporting.

Disclaimer

This project is provided for educational and defensive-security purposes. The risk classification is a rule-based heuristic and should not be treated as definitive proof that an email is safe or malicious. Always perform additional security investigation before making security decisions.